

RESPONSIBLE AI CONSIDERATIONS

EEG-Based Clinical Decision Support Tool

Responsible AI & Clinical Governance Assessment

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1. Purpose

This document outlines the responsible AI principles, ethical considerations, and governance requirements applicable to the proposed EEG-based clinical decision support system. It establishes the ethical framework within which any future development or deployment must operate.

Responsible AI is not an optional layer — in psychiatric healthcare, where decisions directly affect patient wellbeing, dignity, and clinical outcomes, it is a foundational requirement. This document reflects both the current state of the feasibility study and the standards that any clinical deployment would need to meet.

The current project is a proof-of-concept feasibility study and is not intended for autonomous clinical deployment.

2. Responsible AI Principles

2.1 Human Oversight & Clinician Authority

The system is designed strictly as a Clinical Decision Support System (CDSS) - not as an autonomous diagnostic tool. Final diagnostic authority must always remain with the licensed treating psychiatrist. AI-generated risk scores are analytical inputs to clinical judgment, not replacements for it.

- Clinicians must retain the ability to override or disregard model outputs
- AI output must be presented as supplementary information, not primary findings
- Human oversight must be technically enforced through system design and organizationally enforced through governance policy

2.2 Transparency & Explainability

Clinicians and patients have a right to understand how an AI system arrived at its output. A black-box risk score without explanation is clinically and ethically unacceptable in a psychiatric context. The current study provides ANOVA-based feature importance at the aggregate level — deployment requires inference-time explainability at the individual patient level.

- Patient-level prediction explanations required — not just aggregate feature importance
- Clinician-readable analytical summaries must accompany every risk score
- Transparent documentation of model limitations must be included in all outputs

2.3 Fairness & Non-Discrimination

AI systems trained on biased or unrepresentative data risk producing discriminatory outputs — performing well for some demographics while failing others. In psychiatric care, where historically marginalized populations already face diagnostic disparities, this risk is particularly acute.

- Current dataset: 130 subjects, 64% male, primarily 25–45 age range — limited demographic representation
- Formal fairness auditing, subgroup validation, and continuous post-deployment monitoring required

- Model performance must be validated separately across age, sex, ethnicity, and clinical severity subgroups

2.4 Privacy & Data Protection

Patient neurophysiological data is among the most sensitive categories of personal health information. The system must operate within a strict data governance framework at all stages — from research to clinical deployment.

- Anonymized data used in feasibility study — no real patient data collected
- Full HIPAA compliance, end-to-end encryption, and role-based access control required for deployment
- Minimal data retention policies must be defined and enforced

2.5 Safety & Do No Harm

Healthcare AI must meet a higher standard than general AI — errors have direct patient safety consequences. The system must be designed with explicit safeguards against misuse, misinterpretation, and scope creep beyond its intended decision-support function.

- Current model performance is insufficient for independent clinical usage
- Output framing must emphasize the probabilistic and supplementary nature of risk scores
- Clinical accountability must remain clearly defined at the clinician and institution level

2.6 Accountability & Auditability

Every inference the system makes must be traceable, logged, and auditable. When an AI-assisted decision is later reviewed — clinically, legally, or regulatorily — there must be a complete record of what data was used, what the model produced, and how the clinician acted on it.

- Timestamped audit logs required for all inputs, inferences, and outputs
- Clinicians retain responsibility for final decisions
- Healthcare organizations retain governance responsibility for AI system operation

3. Ethical Risk Assessment

RA-01	Diagnostic Overreliance	Impact: High
Risk	Clinicians may over-rely on AI-generated risk scores, reducing the depth of their own clinical assessment — particularly in high-volume or under-resourced settings.	
Mitigation	System design must present AI outputs as supplementary information only. Mandatory clinician confirmation steps required. Training programs must explicitly address overreliance risk.	
RA-02	False Positive Harm	Impact: High
Risk	A patient incorrectly classified as high-risk may experience unnecessary anxiety, stigma, additional procedures, or premature psychiatric labeling.	
Mitigation	Output framing must emphasize the probabilistic nature of risk scores. Clinician training must stress that positive classification requires full clinical corroboration before any action is taken.	
RA-03	False Negative Harm	Impact: High

Risk	A patient with early schizophrenia risk incorrectly classified as low risk may have their concern dismissed, delaying intervention during the critical early window.
Mitigation	The system should be calibrated toward higher sensitivity in clinical use. Clinicians must be trained not to use a negative AI output as a reason to close an assessment without full clinical evaluation.

RA-04	Demographic Bias	Impact: High
Risk	The model was trained on a dataset with limited diversity (64% male, primarily 25–45 age range, single research cohort). Performance on underrepresented groups is unknown and unvalidated.	
Mitigation	Formal bias audit required before deployment. The dataset must be expanded with a representative sampling. Model performance must be validated separately across all demographic subgroups.	

RA-05	Lack of Explainability	Impact: High
Risk	Clinicians who are presented with a risk score they cannot interpret may either blindly trust it or dismiss it entirely — both responses are clinically unsafe.	
Mitigation	Inference-time explainability required. Each patient-level prediction must be accompanied by top contributing EEG features in a clinician-readable format.	

RA-06	Scope Creep	Impact: High
Risk	A tool designed for schizophrenia risk classification may be repurposed or misapplied to other psychiatric conditions for which it was not validated.	
Mitigation	The system must include explicit technical and governance constraints on intended use. Output disclaimers must specify the validated scope. Off-label use must be flagged and prevented at the system level.	

RA-07	Data Privacy Breach	Impact: High
Risk	Patient neurophysiological data is highly sensitive. A breach could expose intimate health information and cause significant harm to patient dignity, trust, and legal standing.	
Mitigation	Full HIPAA compliance, end-to-end encryption, role-based access control, and breach notification protocols are required. Minimal data retention policies are always enforced.	

RA-08	Vulnerable Population Risk	Impact: High
Risk	Patients being evaluated for psychiatric conditions are inherently vulnerable. AI-assisted evaluation adds a layer of opacity that may affect patient trust, informed consent, and the therapeutic relationship.	

Mitigation	A transparent patient consent framework is required. Patients must be clearly informed of the AI system's role, limitations, and the clinician's ultimate authority. Right to opt-out must be preserved.
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4. Current Study — Responsible AI Status

Principle	Status	Notes
Human oversight enforced	✓ Yes	System designed as decision-support only throughout all project artifacts
Transparent limitations documented	✓ Yes	Limitations documented across all project artifacts consistently
Aggregate feature importance provided	✓ Yes	ANOVA SelectKBest — top 100 features identified and documented
Anonymized dataset used	✓ Yes	No real patient data — research dataset only throughout the study
Inference-time explainability	✗ Not yet	Required for deployment — not implemented in the feasibility study
Formal bias audit conducted	✗ Not yet	Dataset limitations documented; formal audit not conducted
Patient consent framework	✗ Not yet	Required for any clinical deployment — not applicable at the feasibility stage
Diverse demographic validation	✗ Not yet	Single cohort, limited demographic range — flagged as deployment requirement
Institutional governance framework	✗ Not yet	Required before any institutional deployment consideration
Post-deployment monitoring plan	✗ Not yet	Required for ongoing responsible AI compliance post-deployment

5. Responsible AI Requirements for Deployment

ID	Requirement	Description	Priority
RAR-01	Explainability Framework	Inference-time explainability producing patient-level contribution scores in a clinician-readable format	High
RAR-02	Bias & Fairness Audit	Formal audit across age, sex, ethnicity, and clinical severity using established fairness metrics with a documented report	High
RAR-03	Representative Dataset	Minimum 500+ subjects with balanced age, sex, and ethnic distribution across multiple clinical sites	High
RAR-04	Patient Consent Framework	Informed consent protocol explaining AI role, limitations, and clinician authority integrated into clinical intake	High
RAR-05	Governance Framework	Institutional AI governance policy covering approval, monitoring, incident reporting, and periodic review	High
RAR-06	Clinician Training Program	Structured training covering AI output interpretation, limitation awareness, overreliance prevention, and clinical judgment	High
RAR-07	Post-Deployment Monitoring	Continuous performance monitoring across demographic subgroups with triggers for review and retraining	High
RAR-08	Output Scope Enforcement	Explicit disclaimers, probabilistic framing, and technical constraints preventing off-label use	High

6. Responsible AI Position Statement

This project intentionally positions AI as an analytical support mechanism — not as an autonomous psychiatric diagnostic system.

The feasibility study demonstrated exploratory machine learning capability. However, responsible healthcare AI deployment depends on significantly broader considerations beyond predictive performance alone.

Safe implementation would require alignment across: clinical validation, regulatory compliance, operational governance, explainability, fairness, and patient-centered healthcare principles.

7. Conclusion

Responsible AI in psychiatric healthcare is not a compliance checkbox — it is the foundation on which clinical trust, patient safety, and ethical deployment are built. This assessment has identified eight distinct ethical risks, each with High impact, and key responsible AI principles against which the current study has been evaluated.

The feasibility study has established responsible framing by positioning AI as decision-support only, documenting limitations transparently, and using anonymized research data throughout. Four principles are confirmed to be implemented. Six remain unimplemented at the level required for clinical deployment.

These limitations reflect the expected gap between a proof-of-concept feasibility study and a production-grade clinical system. Addressing them is the prerequisite for responsible progression from feasibility to deployment.

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